Phase 1 Literature Review

Quantitative Research & Trading Framework

Kartik
Team A · Strategy Design
Industrial Engineering & Operations Research, IIT Bombay

Supervisor: Prof. Sandip Mondal (EE, IIT Bombay)
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Preface

This document presents a structured literature review across the ten themes specified in the project brief. Following the instruction that this should not be a paper-by-paper summary, each theme is organized around **one primary source** (a research paper, book chapter, or — when no good paper exists — an authoritative blog or reference). For each theme I provide: (i) the primary source with justification, (ii) a brief conceptual summary establishing the foundation, and (iii) the dependencies linking this theme to others.

The review culminates in a comparison matrix across strategy families and a dependency map of the factors discussed, followed by a proposed direction for Phase 2.

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1 Classical Strategies

Primary source: Asness, C., Moskowitz, T., Pedersen, L. (2013). "Value and Momentum Everywhere." *Journal of Finance*, 68(3), 929–985.
Why this one: This is the most rigorous out-of-sample validation of the two oldest classical strategy families — value and momentum — across eight asset classes and two decades. Unlike the original Jegadeesh–Titman (1993) and Fama–French (1993) papers, which establish each effect in isolation, Asness et al. examine both jointly across global equities, currencies, country indices, government bonds, and commodities. They demonstrate that momentum and value (i) work globally, (ii) are negatively correlated with each other (forming a natural diversifying pair), and (iii) share common risk factors across asset classes. The paper is the foundational reference for understanding which classical anomalies generalize beyond US equities.
Access: Free at aqr.com (AQR Insights). SSRN abstract id 1363476.

1.1 Conceptual summary

Classical systematic strategies fall into four families: **mean reversion**, **trend following / momentum**, **statistical arbitrage**, and **technical indicator**-based strategies. The first three correspond to distinct hypotheses about how prices behave: prices revert to a mean (mean reversion), prices exhibit persistence (momentum), or linear combinations of prices revert even when individual prices do not (statistical arbitrage).

Mean reversion bets on overshoots. The strategy is short-volatility in disguise: it earns small gains during quiet periods and suffers occasional large losses during trends. It is most reliable when applied to *residuals* (price minus a fitted component), not raw prices, because residuals lack the trend component that breaks the assumption.

Momentum comes in two forms. *Cross-sectional momentum* ranks assets by recent returns and buys winners while shorting losers — Jegadeesh and Titman (1993) established this on US equities with a 12-month formation period skipping the most recent month. *Time-series momentum* (Moskowitz, Ooi, Pedersen 2012) operates on each asset individually, going long if recent returns are positive, regardless of cross-sectional rank. Time-series momentum has been validated across more than fifty futures markets over a century and is the foundation of the managed futures (CTA) industry.

Statistical arbitrage extends mean reversion to baskets of assets. Pairs trading (Gatev, Goetzmann, Rouwenhorst 2006) finds two cointegrated stocks and trades deviations from their historical ratio. Modern PCA-residual statarb (Avellaneda & Lee 2010) decomposes returns into systematic factor components and idiosyncratic residuals, then trades mean reversion of the residuals. The latter is far more scalable and is the dominant approach today.

Technical indicators — moving average crossovers, RSI, MACD, Bollinger Bands — are nonlinear combinations of price and momentum. They are best understood as features within strategies rather than as strategies in themselves. Lo, Mamaysky and Wang (2000) tested chart patterns rigorously and found some weak statistical content but nothing approaching the discretionary trading folklore.

1.2 Where each works and fails

- **Mean reversion**: works in range-bound, low-volatility regimes with high-liquidity instruments. Fails during trends or regime breaks.
- **Cross-sectional momentum**: works in trending, low-volatility equity markets. Crashes after bear-to-bull turning points (e.g., 2009 momentum crash).

- **Time-series momentum**: works across asset classes during sustained directional moves. Whipsaws in choppy markets.
- **Statistical arbitrage**: works when factor structure is stable. Fails when cointegrations break (e.g., August 2007 "quant quake" — see Khandani & Lo).

1.3 Dependencies

Classical strategies' performance depends critically on **regime** (Section 7), **transaction cost structure** (Section 3), and **time horizon** (Section 8). The same momentum signal can produce a 1.2 Sharpe at monthly horizons and a −0.3 Sharpe at intraday horizons after costs.

2 Pattern Detection

Primary source: Cartea, Á., Donnelly, R., Jaimungal, S. (2018). "Enhancing Trading Strategies with Order Book Signals." *Applied Mathematical Finance*, 25(1), 1–35.
Why this one: Of the four pattern types listed in the brief — order book imbalances, candlestick patterns, time series motifs, and sentiment patterns — order book imbalance (OBI) is the only one with consistent empirical support across studies. This paper is the canonical reference. Using NASDAQ Level-II data, the authors define a volume-imbalance measure and demonstrate empirically that it predicts both the sign of the next market order and short-term price changes. They then derive an optimal trading strategy based on this signal using stochastic control. The paper is methodologically rigorous, uses out-of-sample testing, and addresses the central question the brief poses: whether patterns are statistically significant and stable across periods.
Access: Free at SSRN id 2668277. Author Jaimungal posts all papers free at his university page.

2.1 Conceptual summary

Pattern detection asks whether recurring structures in market data are exploitable. The central question is statistical: any sufficiently long time series contains patterns by chance, so the test is whether a pattern (i) generalizes out-of-sample and (ii) carries economic significance after costs.

Order book imbalance is the strongest documented short-horizon pattern. The intuition: if visible buy-side liquidity at the top of the book substantially exceeds sell-side liquidity, the ask is more likely to be lifted than the bid is to be hit, so price drifts upward. The signal decays in seconds to minutes; above five minutes it carries no information.

Candlestick patterns (doji, hammer, engulfing, head-and-shoulders) are popular among discretionary traders. Lo, Mamaysky and Wang (2000) found some weak statistical content but no patterns that survive transaction costs as standalone strategies. They are best used as ensemble features.

Time series motif discovery searches for recurring subsequences in long time series. The Matrix Profile algorithm (Mueen, Keogh et al.) is the efficient modern tool. Applied to finance, motifs are useful as features rather than standalone signals — noise dominates the signal at most horizons.

Sentiment-driven patterns extract textual sentiment from news, social media, or earnings calls to predict short-term returns. Loughran and McDonald (2011) built the standard finance-specific dictionary. Modern LLM-based variants outperform dictionary methods on subtle text. Sentiment is most useful as one feature among many.

Regime-specific recurring behavior ties this theme to regime detection (Section 7). Some patterns appear only in specific regimes; unconditional analysis averages across mutually contradictory behaviors.

2.2 The central question: significance vs noise

Three tests separate real patterns from noise:

- **Out-of-sample stability**: does the pattern hold in a holdout period?
- **Multiple-testing correction**: if many patterns were searched, the best by chance will look significant.
- **Economic intuition**: is there a reason the pattern should exist?

The third test is the most important and the most underused.

2.3 Dependencies

Pattern detection depends on **data resolution** — OBI requires LOB data; sentiment requires news feeds; motif discovery requires sufficient sample. Pattern stability depends on **regime** (Section 7). Pattern significance must be evaluated using the methods in Section 10.

3 Market Structure

Primary source: Cartea, Á., Jaimungal, S., Penalva, J. (2015). *Algorithmic and High-Frequency Trading*, Cambridge University Press. Chapters 1–2 (microstructure primer) and Chapter 10 (market making).
Why this one: This is the standard graduate reference for market microstructure in quantitative finance. Chapters 1 and 2 are an unusually clear introduction to limit order books, bid-ask spreads, price impact, and the mathematical primitives of microstructure. Chapter 10 derives the Avellaneda–Stoikov market-making model from first principles. Selecting a textbook chapter rather than a paper for this theme is deliberate: microstructure is a body of cumulative results, not a single discovery, and Cartea–Jaimungal–Penalva synthesizes the field at the level of detail needed to read the subsequent literature.
Access: Cambridge University Press. Available in the IITB library. The mathematical primer (Section 4 of Chapter 1) is also reproduced in author Jaimungal's lecture notes at his teaching page.

3.1 Conceptual summary

Market structure is the set of mechanisms by which prices form. **Microstructure** refers to the second-by-second mechanics: how orders interact in the limit order book, how spreads emerge, how trades impact prices. **Macro-structure** refers to regime-defining events: central bank decisions, geopolitical shocks, earnings announcements.

A **limit order book (LOB)** aggregates passive orders waiting to be matched, divided into a bid side (buyers) and an ask side (sellers). *Limit orders* provide liquidity by sitting in the book; *market orders* consume liquidity by executing against existing limit orders. The *bid-ask spread* is the gap between the best bid and best ask; the *mid-price* is their average.

Why spreads exist (Glosten & Milgrom 1985): market makers face adverse selection from traders who may know something they don't. To survive, market makers widen the spread enough that

profits from uninformed flow exceed losses to informed flow. Spreads are therefore not transaction costs but rather adverse-selection insurance premia.

Price impact (Kyle 1985): large trades move prices. The Kyle model proposes $\Delta P = \lambda \cdot Q$ where λ measures market depth and Q is signed order flow. Empirically, impact scales as approximately the *square root* of trade size — the "square root law of market impact" is one of the most robust empirical regularities in finance (Bouchaud and collaborators).

Market making (Avellaneda & Stoikov 2008): the optimal market maker quotes bid and ask prices that depend on the mid-price, current inventory, time horizon, and volatility. Asymmetric quotes encourage inventory-reducing trades. This model is the foundation for essentially every academic market-making study.

Optimal execution (Almgren & Chriss 2000): liquidating a large position requires balancing *impact cost* (spread + market impact, minimized by slow execution) against *risk cost* (price variance, minimized by fast execution). The Almgren–Chriss solution underlies TWAP, VWAP, and Implementation Shortfall execution algorithms.

3.2 Macro-structure

Macroeconomic announcements (central bank rate decisions, employment reports, GDP releases) produce predictable volatility spikes. Cross-asset correlations collapse toward +1 during crises, breaking diversification at the moment it is most needed. These facts are not directly tradable in a systematic sense but shape risk-management overlays for almost every strategy.

3.3 Dependencies

Microstructure determines which strategies can survive. Sub-second strategies require low latency and rich LOB data; daily strategies tolerate wider spreads but must still account for them in cost models (Section 10). The structure of the market a strategy will live in must be understood *before* the strategy is designed.

4 Machine Learning & Deep Learning Applications

Primary source: López de Prado, M. (2018). *Advances in Financial Machine Learning.* Wiley. Chapters 1–3 (data structures, labeling, sample weights) and Chapter 7 (cross-validation).
Why this one: Most ML-in-finance papers contain methodological errors that invalidate their results — look-ahead bias, label leakage, plain k-fold cross-validation on serially correlated data, and selection bias from undisclosed multiple testing. López de Prado's book is the only systematic treatment of these problems. The four chapters above teach: how to structure financial data so it is ML-amenable (Chapter 1), how to construct meaningful labels via the triple-barrier method (Chapter 3), how to detect and remove label leakage (Chapter 7), and how to weight samples to reflect their information content (Chapter 4). Selecting a book over a paper is justified here because the field has no single canonical paper — but it does have one canonical critique, and this book contains it.
Access: Wiley. Available at the IITB library. The author's free lecture deck series on SSRN (abstract id 3270329 and sibling IDs) covers the same material.

4.1 Conceptual summary

Standard ML methods perform poorly on naive applications to financial data because financial data violates the assumptions ML methods are built on:

- **Non-stationarity**: the data-generating process changes over time.

- **Low signal-to-noise**: 51% directional accuracy after costs is excellent.
- **Serial correlation**: plain k-fold cross-validation leaks information between folds.
- **Multiple testing**: trying many model variants inflates the apparent performance of the best.

4.1.1 Where ML is genuinely useful

- **Position sizing on classical signals** — Lim, Zohren, Roberts (2019) use DNNs to size positions on top of a time-series momentum signal, improving Sharpe meaningfully without trying to predict prices directly.
- **Meta-labeling** — a primary rule generates trade ideas, a secondary ML model predicts which ideas will be profitable. This decomposition turns out to outperform direct price prediction (López de Prado Chapter 3, 5).
- **Regime classification** — ML can identify regimes (trending, ranging, high-vol) for use in conditional strategies.
- **Stock screening / cross-sectional ranking** — predicting relative returns is easier than predicting absolute returns and is what large quant funds actually do.
- **Microstructure feature extraction from raw LOB** — CNN-based models such as DeepLOB (Zhang, Zohren, Roberts 2019) capture patterns classical methods miss.

4.1.2 Model families

On tabular financial features, gradient boosting (XGBoost, LightGBM) is the default and typically outperforms deep learning. Random forests are a robust baseline. Sequence models (LSTMs, Transformers) help when the input is genuinely sequential (raw LOB, tick data) or multi-asset; they are not magic on OHLCV data. Reinforcement learning is appropriate for execution and sizing problems framed as sequential decisions.

4.1.3 Cross-validation that does not lie

Plain k-fold CV is inappropriate for financial data. The two leakages are (i) overlapping prediction windows and (ii) serial correlation across folds. The correct approaches are **walk-forward analysis** (train on past, test on future, roll forward) and **purged k-fold CV with embargo** (López de Prado Chapter 7), which removes training samples whose labels overlap the test window plus an embargo gap.

4.2 Dependencies

ML signals must be translated into trades (Section 5), evaluated honestly (Section 10), and conditioned on regime (Section 7). An ML model is a component of a strategy, never a strategy in itself.

5 Strategy Formulation

Primary source: López de Prado, M. (2018). *Advances in Financial Machine Learning.* Wiley. Chapter 3 ("Labeling") and Chapter 5 ("Meta-Labeling").
Why this one: The brief lists seven mappings from model output to strategy: classification → long/short/flat, regression → position sizing, ranking → long-top-k, regime → strategy switch, volatility → risk adjustment, anomaly → filter, ensemble → meta-strategy. These two chapters formalize the entire mapping. The triple-barrier method (Chapter 3) defines labels in terms of realistic entry/exit logic rather than next-day returns. Meta-labeling (Chapter 5) is the two-stage decomposition that turns "predict the price" into the more tractable "predict whether this trade will work," enabling position sizing from a binary classifier. This is the most directly relevant chapter pair in the entire AFML book for the prof's stated theme.
Access: Wiley. IITB library. Chapter summaries available in López de Prado's free SSRN lecture decks.

5.1 Conceptual summary

A predictive signal is one component of a strategy. To form a working trading rule, the signal must be combined with: an **entry/exit rule**, **position sizing**, **regime conditioning**, and a **risk overlay**.

The Fundamental Law of Active Management (Grinold & Kahn): Information Ratio ≈ Information Coefficient $\times\sqrt{Breadth}$, where IC is the correlation between a forecast and its outcome and Breadth is the number of independent bets per year. The law explains why cross-sectional equity strategies (thousands of bets per year) can produce high Sharpe ratios with weak per-bet predictions, while single-asset directional strategies (few bets per year) require very strong predictions.

Position sizing options in order of robustness: equal-weight (simplest, but inappropriate when asset volatilities differ); volatility scaling (Carver's preferred approach, where each position contributes equal risk); Kelly-style optimal sizing (theoretically maximal long-run growth, but practically too aggressive — drawdowns are unacceptable, so half-Kelly or quarter-Kelly is more common).

Risk overlays include stop losses, position limits, drawdown breaks, volatility targeting, and correlation constraints. These layers protect the portfolio when the underlying signal is wrong.

5.2 Block-diagram view of a strategy

A full strategy spec is a directed pipeline:

> *[Raw data] → [Cleaning + alignment] → [Feature construction] → [Signal generation] → [Regime filter] → [Position sizing] → [Risk overlay] → [Order generation] → [Execution simulation with costs and slippage] → [P&L and metrics].*

Each arrow carries a defined data flow, frequency, and latency. Most look-ahead bugs hide in the feature-construction arrow; most backtest-vs-live gaps live in the execution-simulation arrow.

5.3 Dependencies

Strategy formulation depends on the signal source (Sections 1, 4), the cost structure of the instrument (Section 9), and the regime (Section 7). Evaluation (Section 10) measures the strategy as a whole, not the signal in isolation.

6 Stock Selection

Primary source: Fama, E., French, K. (2015). "A Five-Factor Asset Pricing Model." *Journal of Financial Economics*, 116(1), 1–22.
Why this one: Factor investing is the dominant paradigm for stock selection, and Fama–French is the foundational framework. The 2015 five-factor extension adds profitability and investment factors to the original three (market, size, value), capturing what later became known as the quality dimension. Selecting the 2015 paper rather than the 1993 original is deliberate: it is the current academic standard, it incorporates the dimensions practitioners actually use, and it sets up the critical question of factor proliferation that Harvey, Liu and Zhu (2016) later answered. Together with the practitioner critique in AQR's "Fact, Fiction and Value Investing" (Asness 2015) — recommended secondary reading — this paper covers the modern factor landscape.
Access: Free on Kenneth French's data library at Dartmouth (French Data Library). Original paper free via SSRN id 2287202.

6.1 Conceptual summary

Stock selection answers: *which* stocks to include in long or short positions. The dominant paradigm is **factor investing**.

Fama and French (1993) introduced the three-factor model: market, size (small minus big, SMB), and value (high minus low book-to-market, HML). The 2015 five-factor extension adds profitability (robust minus weak, RMW) and investment (conservative minus aggressive, CMA). Together with momentum (Carhart 1997, building on Jegadeesh & Titman) and low-volatility (Frazzini & Pedersen 2014, "Betting Against Beta"), these form the core factor set used in industry.

How factor strategies work: rank the universe each rebalancing period by the factor of interest, go long the top decile and short the bottom decile, rebalance monthly or quarterly. The strategy is dollar-neutral. Real-world implementations combine multiple factors using risk-budgeting.

The multiple-testing critique: Harvey, Liu and Zhu (2016) showed that more than 300 "factors" have been published, most of which fail standard multiple-testing corrections. The five to six factors above are those that have replicated globally and survived out-of-sample. Be skeptical of any new factor.

Beyond factors: clustering (hierarchical, k-means, network-based) groups stocks by similar behavior — useful for risk decomposition, pairs identification, and anomaly detection. López de Prado's Hierarchical Risk Parity (HRP) uses clustering to build portfolios more robust to small-sample noise than mean-variance optimization. Earnings-based screens (PEAD — post-earnings announcement drift, profitability, accruals) form fundamental momentum strategies. Sentiment-based reranking adds modest predictive value as a tilt rather than a standalone signal.

6.2 Practical point: universe definition

Before any strategy, the universe must be defined using point-in-time membership (the constituents as of each rebalancing date, not today's constituents). Failing this introduces **survivorship bias**: failed companies disappear from the dataset, inflating backtested returns.

6.3 Dependencies

Stock selection feeds into strategy formulation (Section 5). Factor performance is regime-dependent (Section 7); the value factor's recent underperformance is the most prominent example. Universe construction interacts with the instrument-level cost structure (Section 9).

7 Arbitrage, Market Shifts & Regime

Primary source: Shleifer, A., Vishny, R. (1997). "The Limits of Arbitrage." *Journal of Finance*, 52(1), 35–55.
Why this one: Classical theory treats arbitrage as instantaneous and unlimited. This paper, the foundational reference of behavioral and institutional finance, explains why real arbitrages persist despite being publicly known. The argument is institutional: arbitrageurs use borrowed capital, lenders can withdraw funding, clients can redeem at the worst times, and mark-to-market losses can force liquidation before convergence. The paper is mandatory reading for understanding why no strategy — however clever — can be deployed without risk management. Selecting Shleifer–Vishny rather than a regime-detection paper (e.g., Hamilton 1989) for this theme reflects the more important point: the practical question is not just "when does the regime shift?" but "why does my arbitrage still fail even when I'm right?"
Access: Free at SSRN id 7468 and Andrei Shleifer's Harvard page.

7.1 Conceptual summary

Theoretical arbitrage: a risk-free profit from instantaneous price discrepancies. True arbitrages exist only on millisecond timescales and are captured by high-frequency firms with co-location.

Statistical arbitrage: profitable trades in expectation, with risk. Pairs trading, PCA-residual statarb, and merger arbitrage all fall here. The vast majority of what practitioners call "arbitrage" is statistical.

Risk arbitrage: trades that profit if a specific event occurs. Merger arbitrage (long target, short acquirer at deal announcement); convertible arbitrage; SPAC arbitrage. These carry event risk and capacity constraints.

The limits of arbitrage (Shleifer & Vishny 1997): even when an arbitrage opportunity exists in theory, real-world frictions prevent its full exploitation. Arbitrageurs face capital constraints (collateral and margin), agency problems (client redemptions force liquidation at the worst time), time-horizon mismatch (mispricings may take years to correct), and the risk that spreads widen further before converging. This paper explains why every real strategy — no matter how mathematically clean — eventually has a bad period.

7.2 Regime detection

Markets alternate between regimes with persistent characteristics. The classical reference for regime detection is Hamilton (1989) "A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle," which introduced Markov regime-switching models. Modern approaches include hidden Markov models (with EM-based estimation), change-point detection, and supervised ML on macro/microstructure features. The cheapest and often sufficient approach is **rule-based volatility regimes**: position size inverse to rolling realized volatility implements much of the benefit of explicit regime detection without the estimation noise.

7.3 Dependencies

Regime conditioning enters every strategy in the document. A momentum strategy in a trending regime is a different strategy from a momentum strategy in a ranging regime. The Limits of Arbitrage paper underwrites the necessity of risk management (Section 5) and informs the evaluation framework (Section 10).

8 Time Horizons

Primary source: Lo, A. (2004). "The Adaptive Markets Hypothesis: Market Efficiency from an Evolutionary Perspective." *Journal of Portfolio Management*, 30(5), 15–29.
Why this one: Time-horizon questions are typically not addressed by a single canonical paper because every strategy paper specifies its own horizon. Lo's Adaptive Markets Hypothesis provides the unifying framework: markets are efficient at horizons where many participants compete with similar tools, and inefficient at horizons where they don't. The set of inefficient horizons shifts over time as technology and participants change. The paper is the conceptual lens through which the horizon question should be analyzed; it explains why daily statistical arbitrage that was profitable in 1995 is now largely captured by HFT firms, while weekly factor strategies remain durable.
Access: Free at SSRN id 728864.

8.1 Conceptual summary

Different strategies are viable at different time horizons because the underlying inefficiencies, data, and frictions differ. The horizon spectrum:

- **Sub-millisecond to milliseconds (HFT)**: latency arbitrage, exchange arbitrage. Requires co-location, FPGA hardware, and direct exchange connectivity. Not accessible to academic projects.
- **Milliseconds to seconds**: market making (Avellaneda & Stoikov), order book imbalance (Cartea–Donnelly–Jaimungal). Requires full LOB data and low-latency infrastructure.
- **Seconds to minutes**: order flow imbalance, news reaction. Requires tick data.
- **Minutes to hours (intraday)**: short-term momentum, intraday mean reversion, event reaction.
- **Daily**: cross-sectional momentum, statistical arbitrage, pairs trading. The classical academic horizon.
- **Weekly**: time-series momentum, swing trading.
- **Monthly+**: factor investing, quality, low-volatility.
- **Quarterly+**: asset allocation, macro factor tilts.

The adaptive markets view: as technology has compressed bid-ask spreads and accelerated information flow, the horizon at which alpha exists has shifted. Daily statistical arbitrage that paid handsomely in the 1990s is now mostly captured at intraday and sub-second horizons. Conversely, weekly-to-monthly factor strategies remain durable because long-horizon capital is patient and competition is less intense.

Multi-horizon strategies diversify across timescales — a sub-second market-making book is nearly uncorrelated with a monthly factor book. This is one of the few sources of genuine diversification in modern markets.

8.2 Horizon, data, and cost coupling

The data resolution available determines the minimum horizon that can be studied. Transaction costs determine the minimum horizon at which profit is possible. The intersection of these two constraints narrows the viable strategy space before any strategy is designed.

8.3 Dependencies

Horizon couples to instruments (Section 9: options spreads prohibit HFT), to data availability (no LOB means no sub-second strategy), and to evaluation (Section 10: shorter horizons accumulate more independent observations, allowing tighter statistical inference).

9 Instrument Types

Primary source: Hull, J. (2018). *Options, Futures and Other Derivatives*, 10th ed. Pearson. Chapters 1–6 (mechanics of futures, options, swaps) and Chapter 13 (Black–Scholes).
Why this one: The four instrument classes — equities, futures, options, commodities — differ in liquidity, leverage, expiry behavior, cost structure, and risk profile. No single research paper covers all four; Hull is the universal practitioner-academic reference. The first six chapters establish how each instrument works in practice (margining, mark-to-market, expiry, settlement) and Chapter 13 introduces Black–Scholes pricing for options. This is the minimum foundation for understanding why the same alpha idea looks different across instruments — why options strategies are mostly volatility strategies, why futures admit time-series momentum but not market making, why equity strategies are stock-selection problems while commodity strategies are trend problems.
Access: Pearson. Available at IITB library. Earlier editions (8th, 9th) are functionally equivalent and freely available online for students.

9.1 Conceptual summary

Equities. Liquidity ranges from billions per day (large caps) to thin and gappy (small caps). Leverage is typically capped at 2–10×. Costs are low for liquid stocks (5–10 bps) and meaningful for small caps. No expiry. Suitable for: factor investing, statistical arbitrage, cross-sectional momentum, pairs trading, single-stock event strategies.

Futures. Excellent liquidity in benchmark contracts (e-mini S&P, Bund, Crude). Margin-based built-in leverage with the risk that losses can exceed margin. Very low costs. Must be rolled before expiry, introducing roll cost and carry effects. Suitable for: time-series momentum (managed futures), calendar spreads, basis trading.

Options. Liquidity is highly uneven: near-the-money front-month options are liquid; far-out-of-the-money or far-expiry options are illiquid. Leverage is embedded via gamma. Wide spreads (especially OTM) prohibit high-frequency strategies. Finite lifetime introduces time decay (theta). *Options trading is mostly volatility trading*: strategies trade implied versus realized volatility (gamma scalping), or implied volatility across strikes (dispersion trading), or term structure of volatility. Equity-style direction prediction in options is generally dominated by simply trading the underlying.

Commodities. Generally high liquidity in major contracts (WTI, Brent, Gold). Special features include physical delivery, storage costs, seasonality, and geopolitical sensitivity. Trends are notably persistent — time-series momentum works well.

9.2 The Indian-market context

Indian markets have several distinguishing features relevant to deployment. NSE F&O is one of the largest derivatives segments globally by volume. The Securities Transaction Tax (STT) is a meaningful cost on Indian trades and is asymmetric across instrument types (different rates for delivery, intraday, and F&O), often dominating other costs in retail-scale strategies. SEBI's algorithmic trading regulation continues to evolve. EOD equity data is freely available via `yfinance` and `NSEpy`; intraday and full LOB data require paid feeds (Zerodha Kite, Trueeye).

9.3 Dependencies

Instrument choice determines the cost structure (Section 10), which determines the minimum viable horizon (Section 8), which constrains the strategy family (Section 1). Options additionally introduce non-linear payoff structures that require option-specific risk management (the Greeks).

10 Evaluation Metrics

Primary source: Bailey, D. H., López de Prado, M. (2014). "The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality." *Journal of Portfolio Management*, 40(5), 94–107.
Why this one: The Sharpe ratio is the most quoted performance metric and one of the most misleading. This paper introduces the Deflated Sharpe Ratio (DSR), which corrects the observed Sharpe for three sources of inflation: non-normality of returns, finite sample size, and selection bias from multiple testing. The intuition is that if many strategies are tested and the best is selected, the apparent Sharpe of that best strategy is upward-biased even when the true Sharpe of every strategy is zero. The DSR deflates the observed Sharpe back to what it would have been under a single test. The paper is the single most important reference for honest strategy evaluation — and the companion paper Bailey, Borwein, López de Prado and Zhu (2014) "Pseudo-Mathematics and Financial Charlatanism" provides an even more accessible treatment of the same idea.
Access: Free at SSRN id 2460551. "Pseudo-Mathematics" is free at David Bailey's page.

10.1 Conceptual summary

Sharpe ratio. Mean excess return divided by standard deviation, annualized. Quoted everywhere; misleading on its own because it (i) assumes Gaussian returns, (ii) is symmetric (treats upside and downside volatility identically), (iii) ignores drawdowns, (iv) does not correct for multiple testing.

Better metrics to report alongside Sharpe:

- *Sortino ratio* — penalizes only downside volatility.
- *Calmar ratio* — annualized return divided by maximum drawdown.
- *Maximum drawdown and drawdown duration* — captures the worst-case investor experience.
- *Hit rate* and *win/loss ratio* — diagnoses many small wins versus few big wins.
- *Turnover* — drives transaction costs and capacity.
- *Capacity* — the AUM at which the strategy's own market impact erodes returns.
- *Probabilistic Sharpe Ratio (PSR) and Deflated Sharpe Ratio (DSR)* — statistical-significance versions of Sharpe.
- *Information Ratio* — excess return over benchmark per unit of tracking error.

ML model metrics for the prediction component (classification: accuracy, precision/recall, F1, AUC, log-loss; regression: RMSE, MAE, Information Coefficient). For cross-sectional alpha models the Information Coefficient (Spearman correlation between forecast and outcome) is the most informative single metric.

10.2 Backtesting hygiene

Before reporting any metric: (i) verify no look-ahead bias — every feature at time t uses only information available at or before t; (ii) verify no survivorship bias — universe membership is point-in-time; (iii) include realistic costs — exchange fees, broker fees, taxes, market impact; (iv) include realistic slippage — buy at ask, sell at bid; (v) use walk-forward validation rather than k-fold; (vi) stress test — check performance in known difficult periods (2008, 2020 March, Indian-market-specific stress periods).

10.3 Dependencies

Evaluation is the lens through which every other theme is judged. A strategy with Sharpe 2 in backtest is unremarkable if Deflated Sharpe is 0.3. A strategy with realistic costs and modest in-sample Sharpe may be a far better deployment candidate than one with optimistic costs and high in-sample Sharpe.

11 Comparison Matrix

Strategy family	Instrument	Frequency	Best regime	Reported Sharpe	Main failure mode
Cross-sectional momentum	Liquid equities	Daily–monthly	Trending, low vol	0.5–1.0	Crashes after turning points
Time-series momentum	Futures, FX	Daily–weekly	Trending	0.8–1.2	Whipsaws in choppy markets
Intraday mean reversion	Liquid equities	Minutes–hours	Range-bound	Variable	Trends, regime shifts
Pairs trading (Gatev)	Equity sectors	Daily	Cointegrating	0.4–1.0	Cointegration breaks
PCA-residual statarb	Liquid equities	Daily	Stable factor structure	1.0–2.0	Factor structure shifts
Market making (Avellaneda–Stoikov)	Liquid LOB instruments	Sub-second	Stable spread	1.5+	Adverse selection, inventory blowup
Order book imbalance	Liquid LOB	Sub-second to minutes	Sufficient depth	1.0–2.0	Spoofing, regime change
Value (Fama–French HML)	Equities	Monthly+	Most regimes	0.3–0.6	Multi-year underperformance
Quality (QMJ)	Equities	Monthly+	Most	0.5–0.8	Crashes rare but severe
Low Vol (Betting Against Beta)	Equities	Monthly+	Most	0.5–1.0	Leverage constraints
TS momentum + DNN sizing (Lim–Zohren–Roberts)	Futures	Daily	Trending	1.2–1.5	DNN overfit
LSTM raw price prediction	Anything	Anything	Anything (claimed)	Often 0 OOS	Overfitting

Reported Sharpe figures are taken from the original papers and should be interpreted as in-sample, pre-cost, and pre-deflation. Live deployment commonly produces Sharpe figures 30–60% below reported values.

12 Dependency Map

The prof's brief emphasizes that the ten themes do not exist in isolation. The following dependencies summarize how the themes interact in determining strategy viability.

1. Horizon → Strategy class

Sub-second horizons force market-making or OBI strategies. Daily horizons favor momentum, mean reversion, and statistical arbitrage. Weekly-and-above horizons favor factor investing.

2. Instrument → Cost structure → Minimum viable horizon

Options have wide spreads that prohibit HFT. Liquid equities at EOD have small enough costs to admit daily strategies. The combination of instrument and cost structure narrows the strategy space before strategy selection begins.

3. Microstructure → Strategy survival

Market making fails in illiquid stocks. Momentum fails in illiquid stocks. Pairs trading fails when cointegrations break. The microstructure of a market determines which strategies can survive, independently of how well the strategy is designed.

4. Macro regime → Factor performance

Value has multi-decade periods of underperformance. Momentum has crashes. Trend-following requires sustained directional moves. Most strategies are regime-conditional.

5. Multiple testing → Apparent alpha

Testing many strategies and selecting the best inflates the apparent Sharpe of the survivor. Without Deflated Sharpe correction, the deployed strategy is statistically indistinguishable from noise.

6. Data frequency → Testable strategies

No LOB data implies no testable market-making strategy. No point-in-time fundamentals implies no credible factor backtest. Data resolution constrains the strategies that can be evaluated honestly.

7. Signal → Strategy → Risk overlay

A signal becomes a strategy only when combined with sizing, regime conditioning, and risk overlays. The same signal produces materially different P&L paths under different formulation choices.

8. Regime detection ↔ Pattern stability

A pattern stable on aggregate may be unstable within regimes. Conditioning on regime may reveal patterns hidden by unconditional analysis — or may dissolve apparent patterns that were artifacts of averaging across regimes.

13 Proposed Direction for Phase 2

Three candidate directions emerge from this review, each with a defensible match to my background (IEOR/optimization, IIT Bombay) and the time and data constraints of the project.

Candidate A: Statistical Arbitrage on Indian Equities (PCA-residual)

Following Avellaneda & Lee (2010), decompose Indian equity returns into PCA factor components and idiosyncratic residuals; trade mean reversion of the residuals on a daily horizon. Plays directly to optimization background. EOD data is free. Provides two clean ML extension paths — meta-labeling the residual signal (López de Prado Chapter 5) or LSTM-based entry timing on residual time series.

Candidate B: Time-series momentum on liquid Indian futures with regime filter

Following Moskowitz, Ooi & Pedersen (2012), implement vol-scaled time-series momentum on NIFTY, BANKNIFTY, USDINR, gold, and crude futures. Layer a simple volatility-regime filter. Few parameters — difficult to overfit. ML extension path via Lim, Zohren & Roberts (2019) for DNN-based position sizing.

Candidate C: Short-term equity or crypto prediction via order book imbalance

Following Cartea, Donnelly & Jaimungal (2018), build OBI features from LOB data (crypto: free; NSE: paid), train a classifier on next-N-second direction, combine with inventory rules. Highest match with Prosperity experience. Strongest ML extension path via DeepLOB (Zhang, Zohren, Roberts 2019).

Current preference

Subject to data-access verification with the team, Candidate A is the strongest match: highest leverage on existing optimization skills, free data, clean extension paths, and an academically interesting failure mode (factor-structure shifts) that aligns with the brief's emphasis on regime dependence. Candidate B is the fallback if cross-sectional data quality on NSE proves to be a problem. Candidate C is preferred only if LOB data access can be arranged.

14 Conclusion

The ten themes covered are not independent. A coherent strategy lives at the intersection of *a viable signal* (Section 1 or 4), *a market that can carry that signal* (Section 3, 9), *a horizon at which costs are bearable* (Section 8), *a regime in which the signal holds* (Section 7), *a formulation that turns prediction into trades* (Section 5), *a stock selection process* where applicable (Section 6), and *an evaluation framework that distinguishes signal from selection bias* (Section 10).

The most important insight from this review — spanning López de Prado, Bouchaud, Shleifer–Vishny, and the deflated Sharpe literature — is that the gap between published results and live performance is dominated not by the strategy's signal but by the discipline of its evaluation and the realism of its risk management. Phase 2 will accordingly emphasize methodological rigor (purged cross-validation, walk-forward analysis, deflated Sharpe reporting, realistic cost modeling) at least as much as signal sophistication.

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